# INSTRUCTION MANUAL

# FOR

# THERMOCOUPLE

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# 1. Outline

When welding both ends of two (2) kind of conductors and giving tempera-ture variation to each end, a thermoelectromotive force generates between the conductors caused by flowing thermocurrent in the closed circuit.

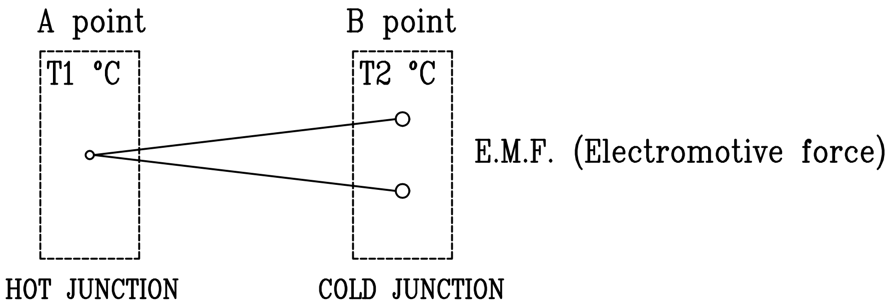


Fig 1.

In Fig 1, value of thermoelectromotive force depends upon the temperature difference between points A and B (T1−T2 °C), that is, the temperature difference between A and B can be given by the value of thermoelectromoti-ve force by means of examinating beforehand the relation between the temperature difference and the thermoelectromotive force.

A thermocouple is the temperature detector using the above theory.

In case of measuring the temperature of point A, it is necessary to add the value of temperature of point B on that which is measured by the value of thermoelectromotive force. For this purpose, a temperature compensa-ting circuit is attached to the circuit.

Meanwhile, in case that the temperature variation at point B is large, the measurement error come into large too.

So, it is considered that the thermoelectromotive force to be measured at the place where the temperature variation is small (point C).

But because, there is temperature difference between point B and C, a compensating lead wire which has approximately the same thermoelectric characteristics as the thermocouple is used for the wiring between points B and C.

In case that the temperature compensation is carried out at point C, usual cables can be used for further wiring. An example of the temperature compensating circuit is shown in Fig 2.

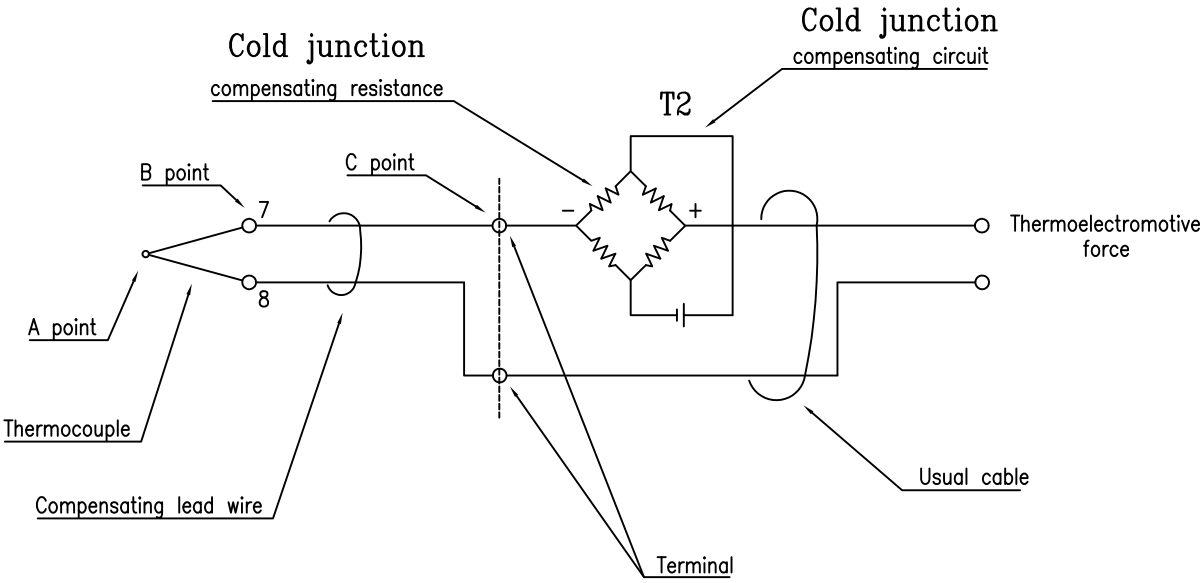


Fig 2.

# 2. Construction

Inner part of the thermocouple consists of the element wire of thermocouple, and outside part consists of the protecting tube.

The protecting tube : This tube is used for protection so that the element wire of thermocouple may not touch the measured thing directly.

# 3. Installation and wiring

Please pay your attention for the followings in case of the installation.

(1) It is necessary to choose the suitable place and the method for the installation which can transmit the temperature correctly and can follow the variation of the temperature as quickly as possible.
Please take care not to bend the tube in view of the construction of the thermocouple.

(2) Please choose the suitable place which has the smallest vibration, the least dirts, and the lowest moisture. Then, please install the Thermocouple at the place which is convenient for the maintenance and the inspection.

(3) In case of the connection of wires, please confirm the wire sign in order to avoid the mis-connection. On the treatment of the end of the wires, please connect the wires firmly by using the compression terminals to avoid the trouble such as the breaking-wire and the short-circuit.

# 4. Maintenance and inspection

## 4-1. Maintenance items and maintenance intervals

| Maintenance / inspection | Initial interval | Normal interval |
|---|---|---|
| * Mechanicial inspection | 1 month | 3 month |
| * Electrical inspection | 1 month | 3 month |
| * Cleaning | 1 month | 1 year |

* Refer to the item 4-6

## 4-2. The condition for opening

(1) Turn off the switch of the electric source.

(2) Do taping around the end of the outer lead-wire which is taken off, in order to avoid the mis-connection.

## 4-3. The order of opening

(1) Take off the cover of the terminal box.

(2) Loosen the terminal screw of the cables.

(3) Loosen the cable gland.

## 4-4. The necessary tools

(1) A plus-driver.

(2) A spanner

(3) A monky spanner

(4) A radio pliers.

(5) A circuit tester

## 4-5. The condition for the maintenance

Prepare the spare parts.

4-6. How to keep good condition

(1) Mechanical inspection

| Inspection points | Inspection method | |
|---|---|---|
| a. Inspection of the looseness of the bolts, plugs and the outer tube. | Examine whether there is any looseness about the tightened part of the bolts, plugs and the outer tube. | |
| b. Inspection of the looseness of the terminals. | Examine whether all the terminals are fixed firmly or not. | |
| c. Cable gland | Examine whether there is any looseness of the lead-in cable gland. | |
| d. Temperature trouble | Examine whether there is any breaking of the gland packing and the wiring. | |

(2) Electrical inspection

| Inspection points | Inspection method | Caution |
|---|---|---|
| a. Inspection of running of the electric source | Examine running of the electric source with a tester. | Whether the breaking wire is or not. |
| b. Insulation resistance test. | Examine the resistance between the terminals and the protection tube with DC 500V tester (more than 5MΩ). | |

(3) Cleaning

| Inspection points | Inspection method | Caution |
|---|---|---|
| Taking away the dirt attached with the protection tube for the high temperature. | After taking off the thermocouple, take away the dirt attached with the insert part. | Avoidance of the error generation and the burning out. |

4-7. Presumption cause or investigation method about the trouble state

| Trouble state | Inspection method | Caution |
|---|---|---|
| Breaking out of the wire | Thermal vibration and shock | Exchange the bad one for the spare part.<br>The pointer of the tester does not move in case of breaking the wire and it shows less than 1MΩ in case of the bad insulation. |
| Bad insulation | a. As for the low temperature, the infiltration of the sea water, fresh water and the oil, due to the badness of the instruments.<br>b. As for the high temperature, the chemical change of the insulation, due to the high temperature. | |
| Short-circuit | Shortage of the distance of the leading-wire | |
| In case that the allowance increases gradually while using. | a. Badness of the indicator.<br>b. Badness of the thermocouple<br>(1) The increase of the thermoelectromotive force, internal resistance.<br>(2) The increase of the dirt attached to the protection tube. | Connect the standard signal generator inst-ead of the thermo-couple and find which of the measuring device of the detector is bad.<br>Exchange for the spare parts.<br>Take off the protection tube and take away the dirt. |
| In case that the allowance increasrs suddenly while using. | a. Abnormality of the measuring device<br>b. Badness of the thermocouple<br>c. Bad connection of the wire, due to the looseness of the screw. | Exchange for the spare parts. |

4-8. Method of repairing

(1) Do in accordance with the reverse order of that of opening.

(2) Concerning the one for the high temperature, the protection against burning out should be carried out around the installation screws of the engine side.

4-9. Confirmation items

(1) Confirm whether the leading-wire is connected correctly.

(2) Confirm the tighteness of the cover.

(3) Confirm the fixed part of the bolts, plugs, etc.

(4) Turn on the switch for the electric source.

5. Selftesting

(1) Visual control and polarity test.

* visual control

check conformity of product with customer's demand

* polarity test

check wires type (color code, magnetism)

| TYPE | COUPLE | COLOR | MAGNETISM |
|---|---|---|---|
| thermocouple | K + | | NO |
| | K - | | YES |
| compensating | KX + | Yellow / Green | NO |
| cord | KX - | Purple / White | YES |